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This is what Allen and I talked about on 2/16/2024.

There is the work of Kogan and Miller which was later followed up by the work of Miller and Sturmfels. In general, we are continuing to think about the story of toric generations of wonderful compactifications of semisimple adjoint groups.

1 Flag variety

We need to start with previous work on the degeneration of the flag variety to the toric variety corresponding to the Gelfand-Cetlin polytope.

1.1 Grobner

The reference for this section is Miller and Sturmfels 2005.

SAGBI stands for subalgebra analogue to Gröbner bases for ideals. Recall the associated graded construction which one can apply to a ring with a given filtration. If we have

$$R = I_0 \supset I_1 \supset \cdots$$

then we can form the associated graded ring

$$\operatorname{gr} R = \bigoplus_{i \geq 0} I_i / I_{i+1}.$$

and if I is a homogeneous ideal in a graded polynomial ring, then $\operatorname{gr} I \subset \operatorname{gr} R$ is an ideal. The point is that in general, when one applies the construction to a prime ideal in a polynomial ring, $\operatorname{gr} I$ is generally prime.

1.2 SAGBI

One way to think about fixing the story is to consider instead subalgebras of our polynomial ring. Now we are thinking about when $\operatorname{gr} I$ is Noetherian.

We don't have many examples of when this happens, but we do have an important one which is in the story of the flag variety.

1.3 Flag variety

We have the following diagram

$$\begin{tikzcd}
\mathrm{GL}_n \arrow[d, "/B"] & \arrow[r] & & M_n \arrow[d, "/B"] \\
\mathrm{GL}_n/B & \mathrm{GL}_n/N \arrow[l, "/T"'] \arrow[r, "i"] & M_n//N := \operatorname{Spec}\mathbb{C}[x_{ij}]^{1\times N} &
\end{tikzcd}$$

The first fundamental theorem of invariant theory tells us what $\mathbb{C}[x_{ij}]^{1\times N}$ is. In particular we are consider rightward column operations except we are not allowed to do the operation of adding a multiple of one column to another or scale columns.
Thus, for $\lambda \in \binom{[n]}{k}$, we have the invariant

$$p_\lambda = \det(\text{first } k \text{ columns and } \lambda \text{ rows})$$

The first fundamental theorem tells us that these p_λ in fact generate all of the invariants. There are quadratic relations which are called the Plucker relations.
Then we have a degeneration which takes p_λ to its diagonal term, and this turns out to be a toric degeneration. The relations in the degernation can be written very nicely

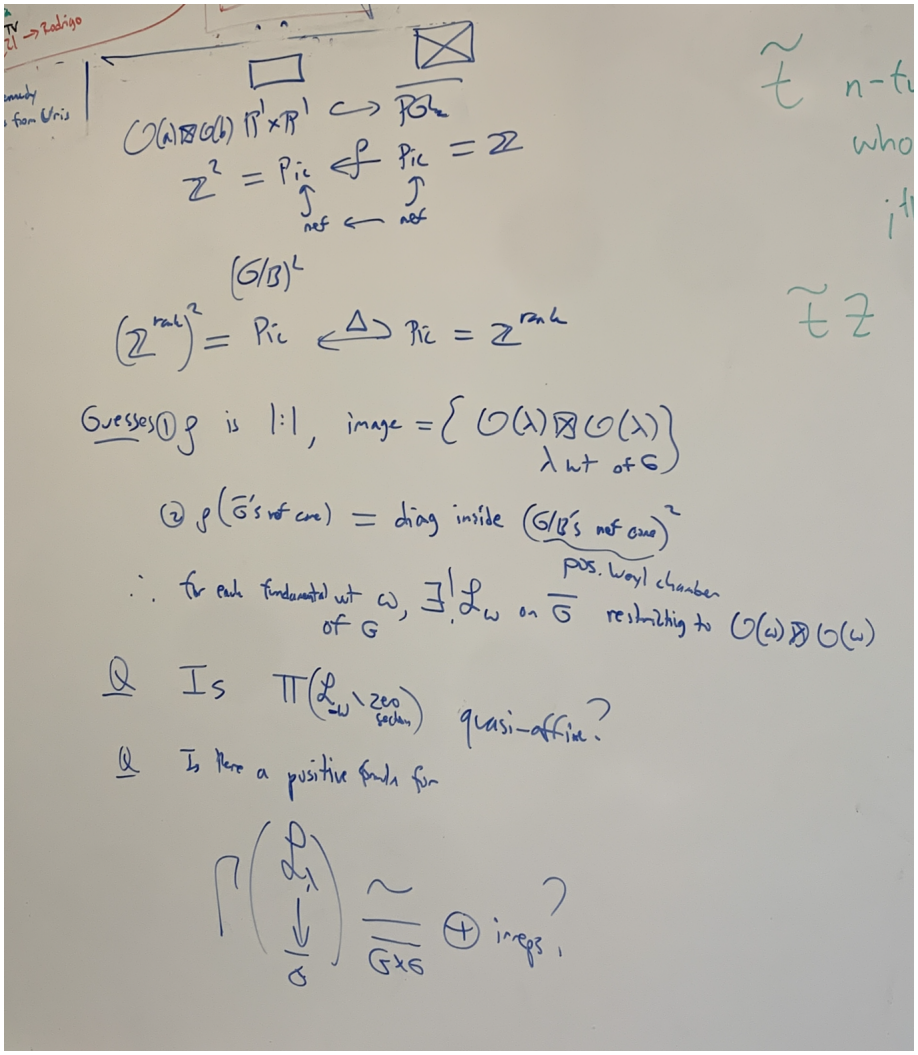
$$p_\lambda p_\mu - p_{\lambda\cup\mu} p_{\lambda\cap\mu} = 0$$

Allen remarks that the key ingredient in this story is the T-bundle $G/B \to G/N$. This is in fact a product of circle bundles, from the generators of the nef cone. Now we are ready to think about the wonderful compactification.

2 Wonderful compactification

The anagalous thing we should think about is the bundle

$$\bigoplus \text{line bundles corresponding to edges of the nef cone} \to \bar{G}$$



3 References

- THE TOTAL COORDINATE RING OF A WONDERFUL VARIETY MICHEL BRION
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